

Theory-Guided LLM Measurement of Legal Delegation and Administrative Discretion

A Human-in-the-Loop Workflow for Reproducible Legal Coding

Draft for discussion

Abstract

Administrative discretion is a latent institutional construct that cannot be reliably recovered from isolated statutory words or a single unconstrained model judgment. Political-economy theory defines discretion as the residual policy latitude available to an administrative actor, jointly determined by the authority delegated by a legislature and the constraints on its exercise. We translate this theoretical structure into a theory-guided large-language-model workflow for coding U.S. financial-regulation laws. The workflow first determines whether qualifying authority is delegated, then separately evaluates the amount of delegated authority and the strength of statutory constraints. These theoretically defined intermediate judgments yield a five-category classification of discretion. We implement the method in Delegation Workflow Studio, a research prototype that exposes prompts, dependencies, evidence, deterministic rules, model outputs, and validation decisions as a versioned research instrument. Human experts intervene at two distinct levels: they revise the general measurement rule and adjudicate individual cases without silently altering the underlying method. The contribution is a general approach to theory-guided AI in which substantive theory determines the structure of computational inference and human review governs its validation and revision.

1 Introduction

Researchers who study political institutions routinely convert complex legal texts into structured measures of authority, constraint, and institutional design (Grimmer and Stewart, 2013). These variables are rarely stated directly in legislation. They are latent constructs inferred from combinations of actors, powers, procedures, limits, and oversight mechanisms. Administrative discretion is a particularly demanding example. It depends on both the amount of policymaking authority transferred to the executive and the legal-administrative constraints governing its use.

The substantive framework originates in political-economy theories of delegation. Legislatures face a trade-off between the informational benefits of administrative expertise and the potential losses associated with bureaucratic drift. They may therefore delegate authority while limiting the permissible scope of agency action through substantive standards, procedures, reporting duties, review provisions, deadlines, and other

controls. Discretion is the residual policy latitude created by these joint institutional choices (Epstein and O’Halloran, 1994, 1999). Later financial-regulation work operationalized this theory into measurable components of delegated authority and constraint (O’Halloran et al., 2015a, 2016b; Groll et al., 2021).

Large language models (LLMs) offer a plausible way to scale legal coding, but prompt-only automation is not a sufficient research design for this construct. A long prompt may contain definitions, exceptions, stage ordering, and output rules, yet the resulting method remains difficult to audit. Existing computational approaches do not fully capture the theoretical structure. Flat classifiers predict a final category from text, while one-shot prompts may include extensive definitions yet obscure how intermediate judgments lead to the final label. A model may return the correct rank for the wrong reason, or the wrong rank because it fails at one of several conceptually distinct stages. It may miss the delegation, conflate implementation with policymaking authority, overstate the legal significance of a procedural requirement, or assign a final category inconsistent with its own intermediate findings.

We address this problem by representing the theory itself as an executable workflow. Delegation is treated as a logically prior gate. Conditional on delegation, the model separately assesses the amount of authority transferred and the strength of constraints on its exercise. The final discretion category follows from combining those dimensions. The sequence is therefore not simply prompt engineering. It is a computational representation of the substantive theory.

This paper develops a theory-guided, human-in-the-loop method for measuring legal delegation and administrative discretion using LLMs. The substantive theory defines the construct and its constituent dimensions. The workflow translates those theoretical relationships into ordered computational judgments, dependencies, and deterministic mappings. The human-in-the-loop design governs the construction, testing, correction, and revision of the resulting measurement instrument. Delegation Workflow Studio implements this method by making each stage, input, output, evidentiary basis, and validation rule visible and versioned. The theory-guided workflow, rather than either the final label or the isolated prompt, is the unit of measurement.

The paper makes four contributions.

1. It translates a well-established political-economy theory of delegation and administrative control into an executable LLM measurement architecture. The ordering and depen-

- dencies of model judgments are derived from the theory rather than selected solely for computational convenience.
2. It reconceptualizes discretion prediction as a structured classification problem. The final ordinal category is derived from separate judgments about delegation, the amount of delegated authority, and the strength of constraints, enabling errors to be pinpointed at the level of the underlying construct.
 3. It develops a human-in-the-loop research design that distinguishes case-level adjudication from method-level revision. Corrections to individual observations do not silently alter the general coding instrument, whereas changes to the theoretical or operational rule require a new version and controlled evaluation.
 4. It implements the method as a reproducible workflow that preserves source policy, prompts, intermediate outputs, evidence, rationales, deterministic branches, model settings, and benchmark comparisons.

2 Theory and Measurement of Delegation and Discretion

2.1 Delegation as an institutional choice

Delegation is the transfer of substantive policymaking authority from a legislature to an executive or administrative actor. It is not equivalent to assigning an actor a task. A statute delegates authority when it permits an administrative actor to make consequential choices that alter policy, determine legal treatment, establish standards, regulate conduct, grant or deny permissions, or otherwise move outcomes away from a legislatively specified status quo.

Legislatures can specify policy directly or leave some decisions to executive actors. Delegation is attractive when administrative actors possess expertise, information, or implementation capacity that legislators lack. It is risky because the agent may use delegated authority in ways that diverge from legislative preferences. The institutional design problem is therefore not only whether to delegate, but how much residual choice to leave after delegation.

2.2 Constraints and residual discretion

Delegation does not by itself determine discretion. Legislatures may combine broad statutory authority with strong procedural or substantive controls, or they may transfer narrower authority with relatively few restrictions. Administrative procedures, approval requirements, formulas, deadlines, consultation duties, reporting obligations, spending limits, review mechanisms, and other legal provisions can restrict the feasible range of agency action.

We therefore define discretion as the residual policy latitude available to an administrative actor after jointly considering the delegation of authority and the constraints governing its use. The published financial-regulation measure formalizes this relationship as:

$$\text{Discretion} = D(1 - C),$$

where D is delegated authority and C is the constraint index. The measure captures a simple theoretical claim: discretion rises when delegated authority is high and legal constraints are weak, and it falls when authority is narrow or constraints strongly structure agency choice.

2.3 Ordinal operationalization

The current LLM workflow translates the continuous theory into a five-category ordinal coding. Rank 0 is determined by the absence of a qualifying delegation. Conditional on delegation, the remaining classes represent the four cells produced by crossing the amount of authority with the strength of constraints. The final rank is therefore derived from theoretically defined component judgments rather than being independently inferred from the text.

Table 1: Theory-derived ordinal operationalization of administrative discretion.

Rank	Authority	Constraints	Interpretation
0	None	N/A	No delegation
1	Low	Strong	Minimal discretion
2	Low	Weak	Limited discretion
3	High	Strong	Bounded discretion
4	High	Weak	Broad discretion

This mapping clarifies why delegation and discretion are not interchangeable. A law may delegate authority while leaving little residual discretion because the statute tightly specifies criteria, procedures, deadlines, or permissible actions. Conversely, a law may confer broad and durable authority with few legal constraints, producing high discretion. The classification task is therefore not simply to detect whether an agency appears in the text. It is to recover the institutional relationship among delegation, authority, constraints, and residual choice.

2.4 Theory-guided computational inference

We use the term *theory-guided AI* to describe a computational design in which substantive theory determines the variables, dependencies, admissible relationships, and evaluation criteria the model uses. Theory is not added after prediction as an interpretive gloss. It structures the inference process itself.

In this application, delegation is logically prior to discretion; authority and constraints are evaluated separately; constraints are assessed by their legal force, scope, proximity to the delegated authority, and consequences for agency choice; and the final category must be consistent with the intermediate findings:

Delegation $\rightarrow$ Authority $\rightarrow$ Constraints $\rightarrow$ Residual discretion.

Figure 1 summarizes the theory-to-classification mapping used in the main workflow.

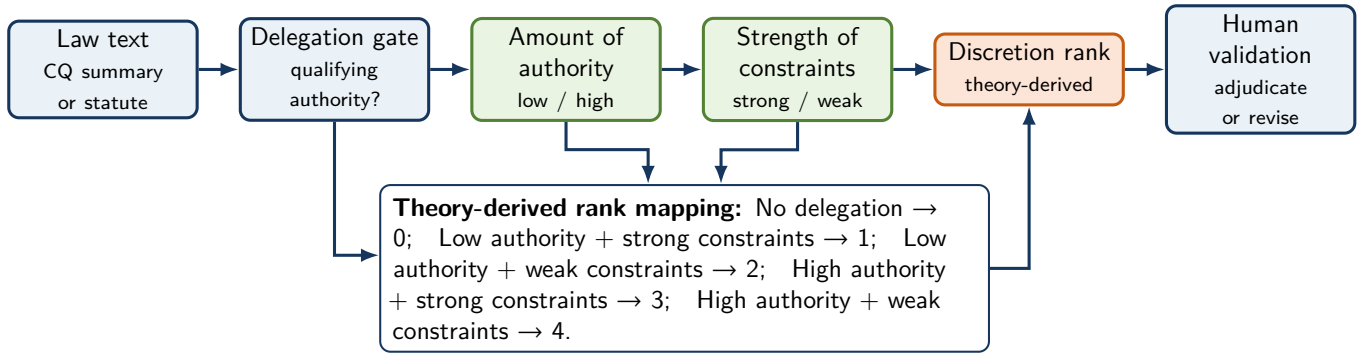


Figure 1: Conceptual theory-guided classification architecture. The model first determines whether a qualifying delegation exists. If no delegation exists, the rank is deterministically 0. Conditional on delegation, the final discretion rank is derived from separate authority and constraint judgments rather than from a single unconstrained classification.

3 Related Work

3.1 From delegation theory to computational measurement

The present study extends a cumulative research program on legislative delegation and administrative discretion. Early formal work conceptualized discretion as the range of policy choices left to an administrative actor and examined how legislatures balance the informational benefits of agency expertise against the risks of bureaucratic drift (Epstein and O’Halloran, 1994). The broader transaction-cost framework later treated statutes as incomplete political contracts, in which legislatures choose whether to specify policy directly or to allocate residual decision rights to executive actors (Epstein and O’Halloran, 1999).

Later empirical work operationalized these concepts using a database of U.S. financial-regulation laws. Delegated authority was measured using provisions that transferred substantive authority, constraints were coded from statutory procedures and controls, and the two were combined into a discretion measure (O’Halloran et al., 2015a, 2016b; Groll et al., 2021). Subsequent computational studies framed the recovery of discretion from legal text as a classification problem and demonstrated the complementarity of expert coding and computational methods (O’Halloran et al., 2015b, 2016a, 2017).

The current paper represents the next step in this progression. Rather than using theory-derived variables only as inputs to a flat classifier, we use the theory to structure the inference architecture itself.

3.2 LLMs as research coders

LLMs have shown promise as low-cost annotators. Gilardi et al. (2023) report that ChatGPT outperformed crowd workers on several text-annotation tasks, while Dai et al. (2023) demonstrate an iterative human–LLM process for thematic analysis. Human-centric work has also emphasized assisting coders rather than treating full automation as the only objective, particularly when rare codes and coder-specific practices matter (Spinoso-Di Piano et al., 2023).

These results do not establish reliability for specialized legal constructs. Legal NLP benchmarks such as LexGLUE and LegalBench emphasize task-specific evaluation and expert-designed categories (Chalkidis et al., 2022; Guha et al., 2023). More directly, Thalken et al. (2023) find that prompted generative models performed poorly when applying a domain-expert codebook to historical Supreme Court opinions, with their strongest results instead coming from an in-domain fine-tuned model. This contrast motivates our conservative position: LLM coding is evaluated against a controlled benchmark and remains subject to expert review.

3.3 Expert supervision and workflow authoring

Weak supervision provides one model for converting expert heuristics into reusable computational signals (Ratner et al., 2017); technology-assisted review in legal discovery likewise demonstrates iterative human feedback over ranked documents (Cormack and Grossman, 2014). Visual prompt-chaining systems are also important predecessors. AI Chains decomposes LLM tasks into modular, controllable steps (Wu et al., 2022b), while PromptChainer provides node-edge authoring, per-step testing, intermediate-output inspection, and debugging for chained prompts (Wu et al., 2022a). We therefore do not claim to invent human-in-the-loop annotation or visual prompt chaining. Our contribution is to integrate those ideas with a substantive institutional measurement model.

4 Theory-Guided Workflow Architecture

The workflow order is grounded in the substantive theory. It is not merely an engineering strategy to reduce prompt length. A law cannot confer administrative discretion without first transferring authority. Once delegation is established, residual discretion depends jointly on the amount of authority transferred and the constraints on its use. The computational architecture therefore mirrors the conceptual structure of the measurement model.

The system evolved through repeated attempts to make the

coding rule both more accurate and more interpretable. Table 2 groups the many prompt files into four methodological phases. The important shift is from treating a prompt as the coding instrument to treating the staged workflow as the coding instrument.

4.1 From theory to ordered computational judgments

The final rank depends on earlier judgments. A model should first determine whether a qualifying delegation exists, then inventory administrative actors and authority, assess the breadth and centrality of that authority, identify constraints, and only then estimate residual leeway. If no delegation exists, the workflow assigns rank 0 deterministically instead of spending another model call on an inapplicable question. If delegation exists, the rank stage receives the earlier values and rationales explicitly.

This dependency structure addresses a recurrent failure of independent-column coding: a later variable such as `discretion_rank` cannot reliably use an earlier delegation decision unless the pipeline stores and forwards it. In the prototype, execution order is computed from the graph rather than assumed from the visual order of columns. Each node records inputs, outputs, status, timing, and, where applicable, evidence and rationale.

4.2 Workflow primitives

The implemented engine supports document-input, LLM, condition, deterministic set-value, validation, output, and rank-descriptor nodes. A workflow definition stores its nodes, edges, typed outputs, and metadata as a database-managed artifact. The distinction between *workflow* and *campaign* is central. A workflow defines the measurement instrument: concepts, prompts, stages, dependencies, branching, and outputs. A campaign or evaluation run applies a selected definition to a document collection under a specified model policy and stores the resulting dataset and traces.

A one-shot prediction collapses the relationships among delegation, authority, constraints, and discretion into an opaque judgment. The workflow makes the measurement assumptions falsifiable because researchers can determine whether a disagreement arose at the delegation gate, the authority assessment, the constraint assessment, or the final mapping.

5 Human Validation and Research Governance

Human expertise enters before, during, and after model execution. Before execution, domain experts define the construct, identify its constituent dimensions, delimit the source universe, specify coding rules, and determine the relationships that the model must preserve. During evaluation, experts distinguish extraction errors from conceptual boundary cases and source-universe mismatches. After evaluation, they decide whether

a disagreement warrants a local case correction, a revision of the operational rule, or reconsideration of the historical benchmark.

Human-in-the-loop design is therefore not a substitute for theory-guided inference. It is the mechanism by which the theoretically structured measurement instrument is governed and validated. Figure 2 presents the broader research process. Solid arrows correspond to implemented or directly supported interactions. The dashed arrow denotes a planned governance step in which selected case-level lessons are deliberately promoted into a new workflow version; corrections do not currently retrain or silently alter the method.

5.1 Method-level revision

At the method level, researchers revise the codebook, source policy, stage prompts, dependencies, output definitions, and deterministic mappings. A method-level revision changes the general measurement rule and should therefore create a new version and controlled rerun before adoption. This protects the comparability of coded observations across documents and model runs.

5.2 Case-level adjudication

For an individual coded law, a researcher can inspect final values and node-by-node execution traces, manually override a selected value and rationale, or request AI re-evaluation with corrective feedback. History distinguishes initial AI output, manual override, and AI re-evaluation. A manual correction is local to that law; it does not automatically change the codebook, the prompt, or other cases.

This separation is scientifically useful. Case correction protects the dataset from a known error, whereas method revision changes the coding rule. The prototype therefore avoids an opaque “learning” process in which corrections silently alter subsequent classifications.

6 Benchmark and Experimental Design

6.1 Benchmark and source discipline

The benchmark consists of the project’s previously published and adjudicated expert coding of financial-regulation laws. We treat these observations as the operational benchmark for replicating measurements, not as infallible legal truth. A disagreement may arise from model error, source mismatch, ambiguity in the operational rule, or error in the historical coding. The historical data were generated through independent coding and multiple levels of review, thereby strengthening the benchmark’s credibility.

The source universe is methodologically important. The benchmark labels were produced from CQ summaries or corresponding descriptions of major provisions. Full statutory text can contain technical or minor delegations absent from those summaries. Consequently, a full-text model may be legally

Table 2: Development from prompt-only classification to a theory-guided workflow research instrument.

Phase	Representation	Methodological change	Observed motivation	Auditability
I	One-shot prompt	Joint binary delegation and 0–4 discretion prediction with rationale and evidence; enforce no-delegation $\Rightarrow$ rank 0.	Establish a first end-to-end coding baseline.	Final rationale, but theory-stage errors remain entangled.
II	Decomposed prompt	Separate label-only and rationale-bearing tasks; then separate delegation, amount/centrality, constraints, constraint categories, and rank.	Isolate where the model fails relative to the theory.	Stage text is visible, but ordering is still embedded in prompt files.
III	Coding-sheet-aligned staged prompt	Map stages to explicit variables; tighten definitions for agency creation, appointment, external audit, sunset provisions, financial-domain scope, and omnibus laws.	Reduce false positives and align outputs with the expert benchmark’s operational rule.	Intermediate variables are explicit but not yet a reusable executable graph.
IV	Versioned workflow	Convert theory stages into typed nodes, upstream dependencies, conditions, deterministic assignments, validations, outputs, reusable templates, execution traces, and published snapshots.	Make the measurement instrument human-editable, testable, and reproducible across documents and models.	Node-level values, rationales, timing, provenance, and branch decisions are inspectable.

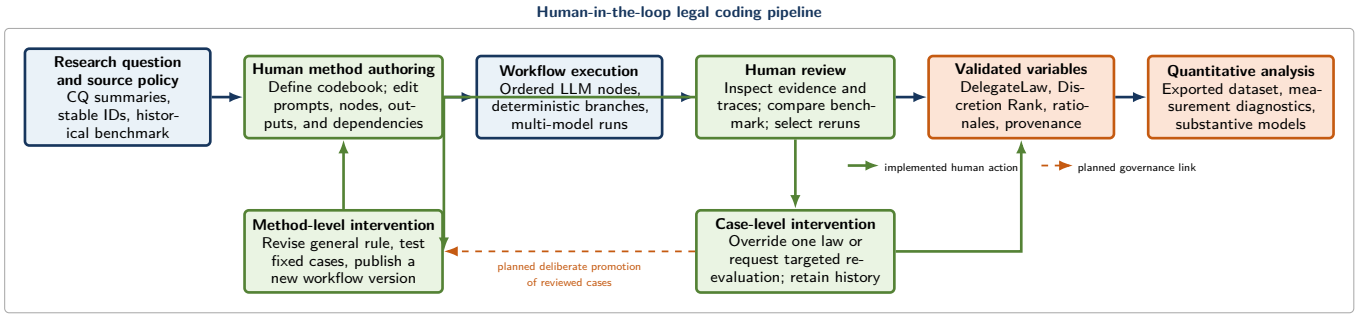


Figure 2: Human-in-the-loop governance around the theory-guided measurement instrument. Human experts define the construct and source universe, validate model outputs, adjudicate cases, and decide when disagreement should trigger a local correction, a new workflow version, or reconsideration of the benchmark.

plausible while disagreeing with the historical label for reasons unrelated to prompt quality. Benchmark runs therefore use the same summary-level information set; full-text runs are labeled exploratory.

The repository currently contains 169 summary files (161 delegation-positive and eight delegation-negative) spanning 1950–2013, which represent the intersection of historical legislative records and available summaries. The broader, unlabeled document database in the prototype extends through 2020 (totaling 412 documents), while the benchmark CSV contains 121 manually coded financial laws spanning 1950–2013.

6.2 Evaluation levels

The empirical evaluation asks whether LLMs can reproduce a theory-based measurement instrument, not merely whether they can generate plausible descriptions of statutes. The evaluation therefore occurs at several levels: whether the model

identifies qualifying delegation, correctly characterizes the amount of authority, correctly characterizes the strength of constraints, produces a final rank consistent with those component judgments, and agrees with the historical benchmark.

Where supported by the current dashboard exports, the evaluation reports delegation-screen accuracy, class-specific recall, final-rank exact accuracy, within-one accuracy, mean absolute error (MAE), class distribution, and law-level disagreements. The next revision should add component-level authority classification, component-level constraint classification, linear and quadratic weighted kappa, and updated v8, v8.2, v8.4, and v9 results once those exports are finalized.

6.3 Controlled prompt and workflow comparison

The testing protocol fixes a development set, holds source documents constant, records prompt and model versions, pre-

serves law identifiers, and compares each output with the benchmark. After each revision, the same laws are rerun and disagreements are reviewed law by law. The protocol explicitly distinguishes recurring errors from isolated mistakes.

The recorded error taxonomy includes: treating any delegation as high discretion; treating minor technical delegation as central; missing constraints; treating rulemaking authority as unconstrained when standards or procedures exist; inventing authority not found in the supplied text; relying on full-text evidence absent from the summary; producing a final rank inconsistent with earlier stages; and confusing delegation with discretion.

The canonical 15-law development set is composed of the 13 delegation-positive summaries available in the summary folder, supplemented by two delegation-negative cases from the broader corpus (PL96-3 and PL92-165) to support class-balanced evaluation.

6.4 Model and workflow comparison

The evaluation dashboard can lock one workflow definition, apply it to the same files across multiple selected models, retain model-keyed values and traces, and report per-variable matches, rank error, token use, cost, timing, and failures. This design separates the coding strategy from the model used to execute it. The campaign evaluation interface can summarize model-level costs, token use, per-strategy benchmark agreement, mapped cells, and variable-level matches.

6.5 Metrics

For the binary delegation task, we report overall accuracy, positive-class recall, negative-class recall, and balanced accuracy. If the historical positive class is delegation, then

$$\text{BalancedAccuracy} = \frac{1}{2} \left(\frac{\text{TP}}{\text{TP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP}} \right).$$

For the ordinal discretion task, the evaluation interface supports exact accuracy, accuracy within one rank, mean absolute error, and a 0–4 confusion matrix. These measures distinguish an adjacent-rank disagreement from a substantively larger error.

7 Preliminary Findings

7.1 Binary delegation

Table 3 reconstructs metrics from persisted local dashboard rows. The benchmark is severely imbalanced: 161 positive cases and only eight negative cases. An always-positive classifier would obtain 95.27% raw accuracy while failing every negative case. Raw accuracy is therefore not an adequate measure of whether prompt refinement improves the underlying coding rule.

Table 3: Preliminary binary delegation results reconstructed from persisted dashboard rows. Positive denotes the benchmark delegation-positive class.

Strategy	N	Acc.	Pos. rec.	Neg. rec.	Bal. acc.
Prompt v7, Stage 1 v4	169	94.67%	96.89%	50.00%	73.45%
Prompt v8	169	91.72%	91.93%	87.50%	89.71%

Model: `gemini-3.1-flash-lite-preview`. Confusion counts (TP/FN/TN/FP) are 156/5/4/4 for v7 and 148/13/7/1 for v8. Runs were executed on June 20, 2026.

The looser v7 strategy yields the better raw accuracy because it preserves positive-class recall. The conservative v8 strategy makes more false-negative errors, but it detects seven of the eight rare negative cases and produces a much higher balanced accuracy. The substantive result is therefore a trade-off, not a monotonic claim that one prompt is “more accurate.” If the research priority is recovering rare non-delegation cases for expert review, v8 is the stronger baseline. If the priority is maximizing delegation recall, v7 is stronger.

7.2 Diagnosing disagreement

The law PL83-577 illustrates why source discipline and intermediate evidence matter. The benchmark codes it as non-delegation with rank 0. An exploratory staged full-text run instead identified Securities and Exchange Commission rule-making and exemption authority and returned delegation with rank 3. This disagreement could reflect an overbroad coding rule, a source-universe difference, or a mismatch between historical summary coding and full statutory detail. A final label alone cannot distinguish these explanations; a trace that records actor, authority, source evidence, and stage decisions can.

This example shows the value of theory-guided decomposition. Researchers can ask whether the disagreement begins at the delegation gate, the authority assessment, the constraint assessment, or the final mapping. The workflow therefore turns a single wrong label into a diagnosable measurement disagreement.

7.3 Discretion rank and model comparison

Early 15-law experiments provide exploratory ordinal results (Table 4). Exact agreement remained modest, reaching 40% in the strongest checked runs, while 73.3% of predictions were within one rank across all four strategies. These small-sample results motivated clearer separation of delegation screening, authority assessment, constraint extraction, and final ranking. They should not be treated as a held-out estimate of final workflow performance.

Table 5 reports the same 15-law development set across six models and three ranking strategies. CASCADE usually gives the strongest exact accuracy, while M9 often improves within-one-rank accuracy or mean absolute error. Because the sample is small, these results are diagnostic rather than a held-out performance estimate.

Table 4: Exploratory Discretion Rank results on 15 laws. All runs used `gemini-3.1-flash-lite-preview`.

Dashboard strategy	Exact	Within 1	MAE
Prompt v3	26.7%	73.3%	1.067
Prompt v3.4	40.0%	73.3%	0.933
Prompt v3.5	40.0%	73.3%	0.933
Prompt v3.6	33.3%	73.3%	1.000

Metrics were reconstructed from persisted dashboard rows matched to the benchmark CSV by public-law number. Campaigns v3.4, v3.5, and v3.6 share the identical prompt text (Prompt Version 7.2); variance in metrics is due to LLM nondeterminism.

Table 5: Cross-model evaluation results on the 15-law development set for CASCADE, M9, and B3 ranking strategies.

Model	Strategy	Exact	W1	MAE	N
G3.5	CASCADE	50.0%	57.1%	0.929	14
	M9	21.4%	71.4%	1.071	14
	B3	28.6%	64.3%	1.143	14
G3.1-lite	CASCADE	46.7%	73.3%	0.867	15
	M9	40.0%	80.0%	0.800	15
	B3	40.0%	60.0%	1.067	15
Qwen3	CASCADE	46.7%	60.0%	1.133	15
	M9	26.7%	73.3%	1.133	15
	B3	26.7%	46.7%	1.533	15
GLM-5.2	CASCADE	40.0%	73.3%	1.000	15
	M9	40.0%	73.3%	1.000	15
	B3	33.3%	60.0%	1.133	15
DeepSeek	CASCADE	40.0%	73.3%	0.933	15
	M9	26.7%	80.0%	1.067	15
	B3	26.7%	40.0%	1.667	15
Hermes	CASCADE	33.3%	53.3%	1.333	15
	M9	33.3%	60.0%	1.333	15
	B3	26.7%	60.0%	1.333	15

W1 = within one rank. G3.5 = `gemini-3.5-flash`; G3.1-lite = `gemini-3.1-flash-lite`; Qwen3 = `qwen/qwen3-235b-a22b-2507`; GLM-5.2 = `z-ai/glm-5.2`; DeepSeek = `deepseek-v4-flash`; Hermes = `nousresearch/hermes-3-llama-3.1-405b`.

In these exploratory runs, the strongest exact-accuracy result is `gemini-3.5-flash` with CASCADE (50.0%), while the lowest MAE is `gemini-3.1-flash-lite` with M9 (0.800).

8 Discussion

8.1 A theory-guided measurement architecture

The principal contribution is a theory-guided measurement architecture for legal text. The workflow is valuable because it exposes prompts and preserves the substantive structure that links delegation, constraints, and residual discretion. It turns a political-economy theory of regulatory discretion into a reproducible computational instrument.

This design is aligned with a quantitative social-science perspective. The goal is not merely to produce plausible legal interpretations. It is to operationalize a concept consistently, compare predictions with a coded benchmark, diagnose systematic disagreements, and preserve enough provenance for another researcher to reproduce the coding process. In that

sense, the workflow plays a role analogous to a versioned coding protocol or measurement instrument.

8.2 What the human contributes

The human remains responsible for defining constructs, choosing the source universe, approving the codebook in the substantive sense, interpreting benchmark disagreements, correcting known errors, and deciding whether a case-level lesson warrants a method-level revision. The LLM contributes scalable extraction, structured intermediate judgments, and candidate classifications. The system contributes orchestration, provenance, comparison, and versioning.

The two human loops should not be conflated. A local override is appropriate when a researcher knows that one case is wrong. A workflow revision is appropriate when the error reveals a general rule. The latter should create a new version and be evaluated on a controlled set before adoption.

8.3 Limitations

Several limitations bound the current evidence. First, the binary benchmark is extremely imbalanced, and prompt development on the same cases risks overfitting. Second, the benchmark is based on summaries; conclusions do not automatically transfer to full statutes. Third, updated component-level authority and constraint metrics have not yet been incorporated into this draft. Fourth, a single formal run per strategy does not establish statistical stability, even if spot reruns return identical outputs. Fifth, ordinal ranks can propagate errors from earlier stages. Sixth, the current study concerns one policy domain and has no external-dataset validation. Seventh, model and provider versions can drift. Finally, some audit and governance capabilities remain incomplete: validation is diagnostic, actor-attributed override logs are limited, and workflow-version approval and pinning are not yet uniform across the product.

8.4 Novelty claim

We do not claim to introduce the theory of delegation, the measurement of administrative discretion, human-in-the-loop annotation, or visual prompt chaining. Each has an established literature. The contribution lies in combining them in a specific way: translating a substantive institutional theory into an executable LLM architecture whose intermediate variables and dependencies correspond to the theory’s constituent concepts; separating generative judgments from deterministic theoretical mappings; distinguishing correction of an observation from revision of the measurement instrument; and retaining the evidence and provenance needed to diagnose disagreement.

9 Replication Workflow

The current prototype can be replicated through the following research-facing sequence:

1. **Prepare the corpus.** Upload CQ summaries or corresponding major-provisions text and retain stable law identifiers. Keep full statutes in a separately labeled exploratory collection.
2. **Create the instrument.** Duplicate the delegation-and-discretion workflow template. Review every stage’s theoretical purpose, prompt, required inputs, typed outputs, and final/internal visibility.
3. **Validate and test.** Run representative positive, negative, and boundary cases. Inspect graph validation, final outputs, and node-level traces.
4. **Freeze a version.** Save a changelog and publish a numbered workflow snapshot before a benchmark run. Linking a campaign copies the workflow’s definition and revision directly to the dashboard, preventing subsequent edits from silently affecting active runs.
5. **Run a controlled evaluation.** Apply the same workflow and source set across selected models or strategies. Record model identifiers, settings, timestamps, failures, token use, and cost.
6. **Compare with the benchmark.** Map benchmark columns, calculate binary class-specific metrics and ordinal rank metrics, and inspect every mismatch.
7. **Intervene explicitly.** Correct a case locally when appropriate. If the disagreement reveals a general rule, revise the workflow separately, publish a new version, and rerun the fixed development set.
8. **Export the dataset.** Preserve final variables alongside the workflow version, model, source universe, and audit history needed for downstream quantitative analysis.

10 Future Work

Immediate empirical work should incorporate the latest v8, v8.2, v8.4, and v9 results, export the live multi-model results, evaluate component-level authority and constraint performance, and repeat runs to characterize model variability. Methodological work should add a held-out evaluation set and, if feasible, independent recoding to estimate intercoder reliability.

Product work should add role-specific governance, workflow-version comparison and pinning, model-aware manual correction in the evaluation dashboard, editable validation rules, and a deliberate mechanism for promoting reviewed cases into a new calibration set or workflow version. External validity should be tested on an additional policy dataset before making claims beyond financial regulation.

11 Conclusion

This paper began with a longstanding measurement problem: how to recover delegated authority and administrative discretion from legal text. Political-economy theory provides a clear conceptual answer. Legislatures may transfer varying amounts of authority to administrative actors and impose varying constraints on their exercise; discretion is the residual policy lati-

tude of those joint choices.

The methodological challenge is to make that theory executable at scale without reducing it to either a black-box classifier or a single opaque prompt. We address this challenge with a theory-guided LLM workflow. Delegation, authority, and constraints are treated as distinct yet interdependent components, and the final category is derived from their theoretically specified relationship. Human experts govern the source universe, codebook, validation process, case adjudication, and instrument revision.

The resulting system does not remove judgment from institutional measurement. Instead, it makes judgment explicit, structured, inspectable, and reproducible.

A Implementation Appendix: Delegation Workflow Studio

A.1 Detailed workflow graph

The main text foregrounds the conceptual theory-to-classification architecture. Figure 3 records the more detailed workflow graph used in the prototype, including the delegation screen, feature extraction nodes, parallel CASCADE/M9/B3 ranking strategies, validation checks, and final output serialization.

A.2 Workflow authoring interface

The Stage Builder exposes source-text policy, evaluation mode, calibration settings, each stage’s purpose and prompt, typed output fields, and whether a field is final or retained only for audit. A live preview shows the compiled prompt. The more general graph editor exposes upstream-field selection, conditional routing, deterministic assignments, validations, and rank-specific criteria. This lets a domain expert revise the operational codebook without editing application source code.

Researchers can create a workflow from a template, duplicate it, import or export its JSON representation, edit its draft, validate it, test it, and publish a numbered snapshot with a changelog and definition hash. Publishing creates an immutable numbered definition snapshot, but the current prototype does not implement a separate professor-approval request or sign-off state. Any authorized editor can publish. Similarly, the current user interface does not yet provide a complete published-version comparison and pinning workflow for all campaign types. We therefore describe publishing as a reproducibility primitive, not as a completed governance protocol.

A.3 Evaluation interface

The campaign evaluation interface can summarize model-level costs, token use, per-strategy benchmark agreement, mapped cells, and variable-level matches. Figure 5 shows a compact product view of this comparison for two selected models.

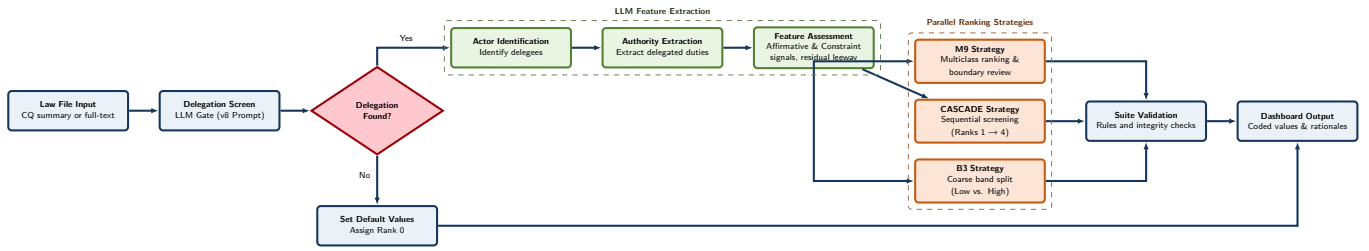


Figure 3: Detailed modular architecture of the Delegation Workflow Studio workflow graph (Version 4 - Detailed). Upstream outputs flow into parallel CASCADE (sequential), M9 (multiclass), and B3 (coarse split) ranking strategies, followed by validation checks and final output serialization.

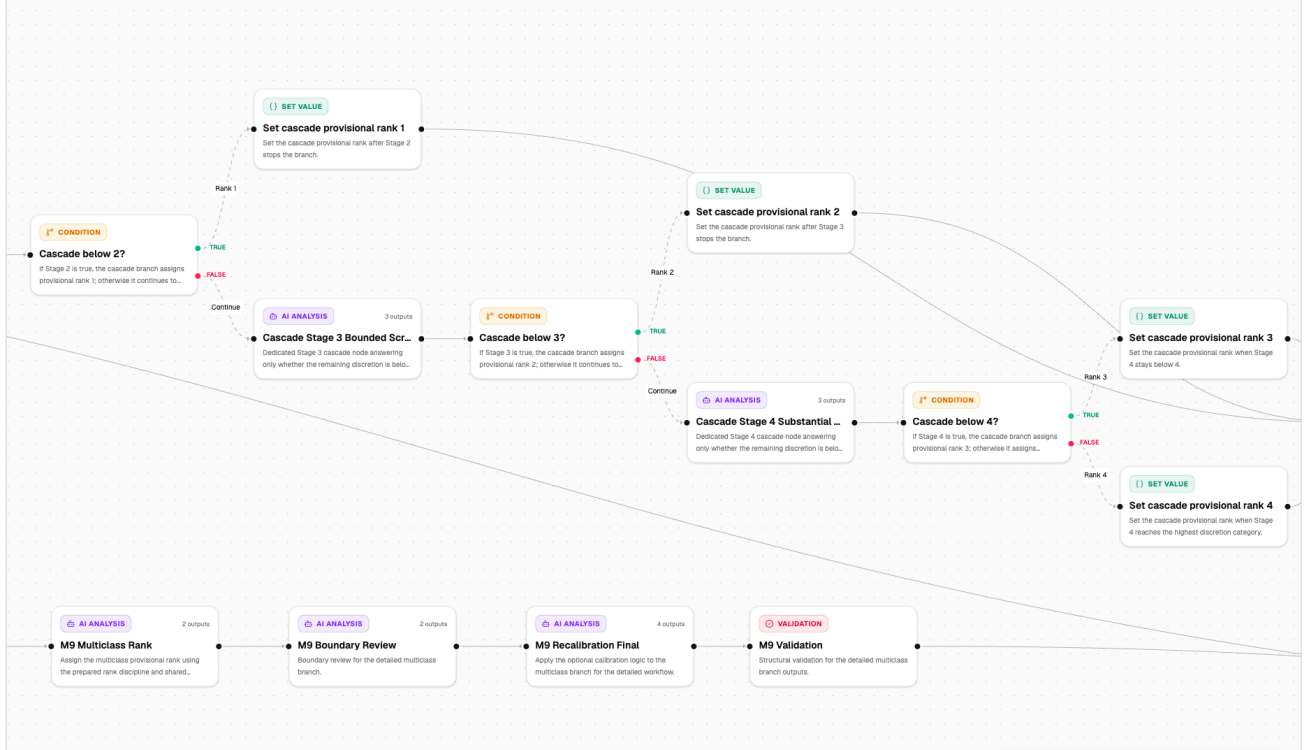


Figure 4: Product view of the workflow graph, showing how the discretion-rank task is decomposed into explicit LLM analysis nodes, condition nodes, deterministic set-value nodes, validation nodes, and connected branches.

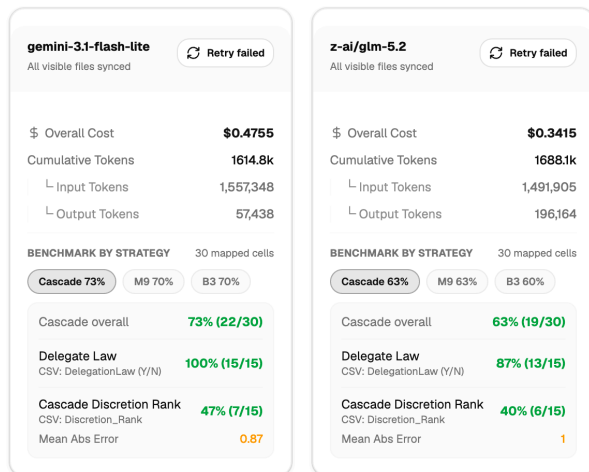


Figure 5: Benchmark-comparison interface showing model-level cost, token counts, strategy-level agreement, and variable-level matches for selected models.

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